

Eva

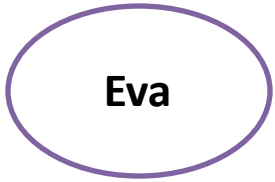
SMART INDIA HACKATHON 2026



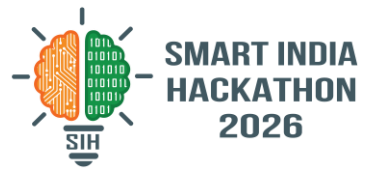
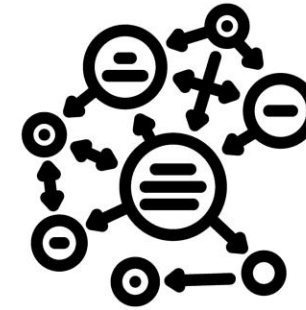
SMART INDIA
HACKATHON
2026

- **Problem Statement ID** – SIH26108
- **Problem Statement Title** - AI-Powered Recommendation Engine for Applicable Indian Standards.
- **Theme** - Smart Automation.
- **Problem Statement Category** – Software.
- **Team ID** -
- **Team Name** - Eva





IDEA TITLE



❑ Proposed Solution :

AI-powered system that analyses a product description and recommends the most relevant Indian Standards automatically.

❑ Addresses The Problem :

Reduces the time and effort required to **manually search through thousands of Indian Standards** and identify applicable ones.

❑ Innovation :

Uses AI and semantic understanding to provide **context-based recommendations**, even when the user doesn't know the exact standard number.

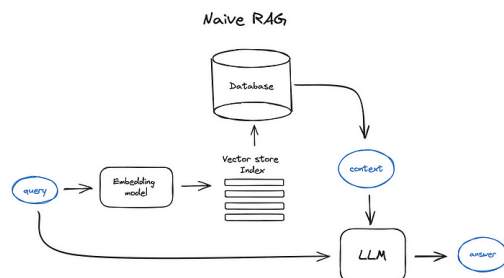
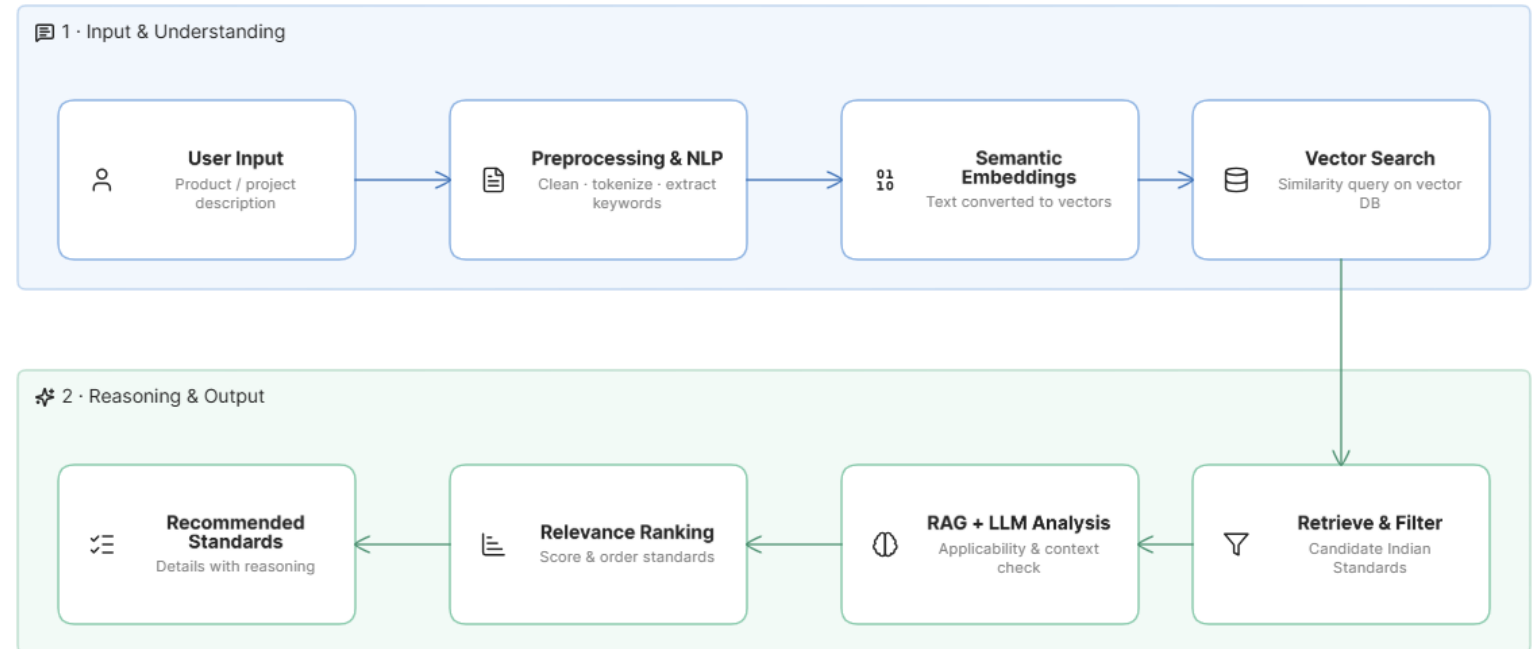


TECHNICAL APPROACH

❑ Technologies :

1. Python
2. React.js
3. Fast-API
4. Olama 3.2 3B LLM
5. Local RAG Pipeline
6. Vector Database
7. Embeddings
8. Multi-Language NLP

❑ Methodology



FEASIBILITY AND VIABILITY

❑ Analysis Of The Feasibility Of The Idea :

Uses scalable AI and web technologies with readily available standards data, making the solution technically and economically feasible.



❑ Potential Challenges And Risks :

Data quality, incomplete standards information, AI-generated inaccuracies, and keeping standards up to date.

❑ Strategies For Overcoming These Challenges :

Use verified BIS sources, regular data updates, RAG-based validation, relevance scoring, and human verification for critical recommendations.



IMPACT AND BENEFITS



❑ Potential impact on the target audience :

Enables industries, manufacturers, engineers and researchers to quickly identify applicable Indian Standards, reducing manual effort and search time.



❑ Benefits of the solution :

Improves compliance, reduces costs and errors, supports faster product development, and promotes standardized and sustainable practices.



RESEARCH AND REFERENCES

- **Bureau of Indian Standards —**
Official Standards Portal
[BIS Standards Portal](#)
- **Know Your Standard —**
Search standards by IS number/keyword
[BIS Know Your Standard](#)
- **BIS —**
Compendium of Indian Standards
[BIS Compendium of Indian Standards](#)
- **RAG for Large Language Models —**
Research Survey
[RAG Research Paper](#)
- **Retrieval-Augmented Generation —**
Research Survey
[RAG Survey](#)
- **Indian Standards —**
Standardization Research Papers
[Bureau of Indian Standards](#)



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